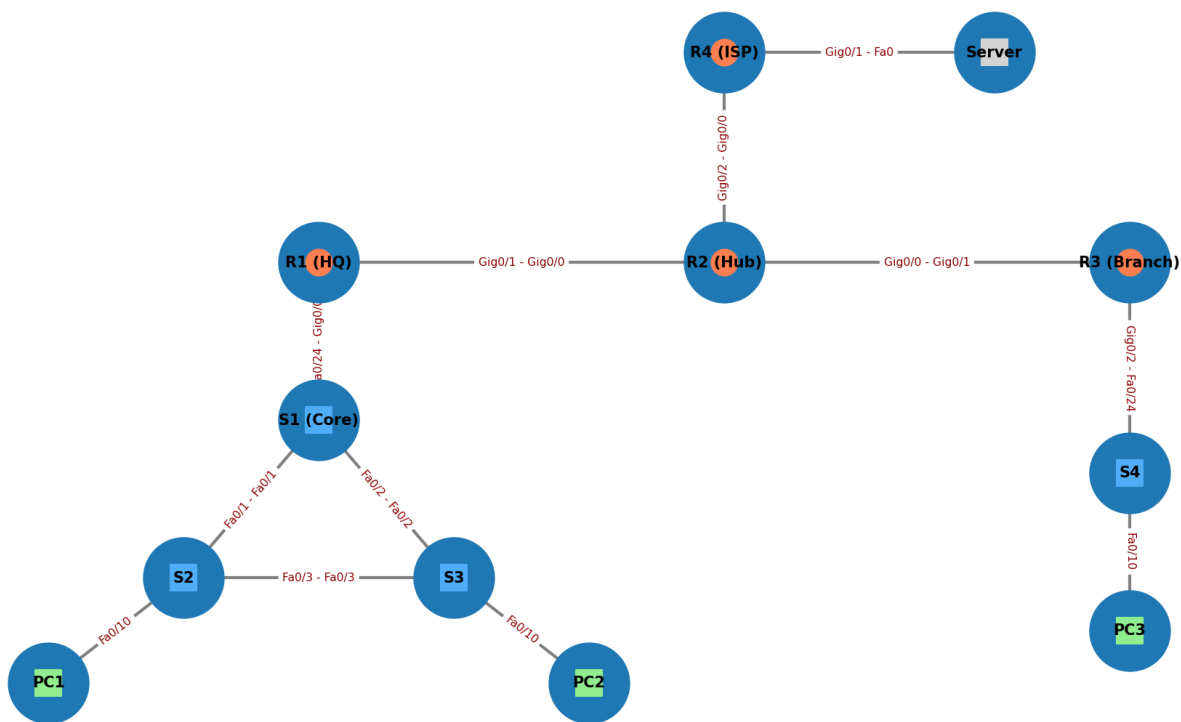


ISPITNI ZADATAK 2: Centrala i Filijala (RIPv2, NAT, STP)

Logička Mrežna Topologija: Centrala i Filijala (Hub & Spoke)



1. Opis Mreže (Hub & Spoke dizajn)

Ova mreža simulira kompaniju sa glavnom centralom (HQ) i jednom izdvojenom filijalom (Branch). Svi podaci ka internetu idu preko centralnog Hub rutera (R2).

- **Centrala (HQ):** Sastoji se od tri sviča (S1, S2, S3) u petlji i rutera R1. Tu se nalaze sektori Prodaja (VLAN 10) i Marketing (VLAN 20).
- **Filijala (Branch):** Sastoji se od rutera R3, sviča S4 i klijenata (VLAN 30).
- **Jezgro i ISP:** R1 i R3 se spajaju na centralni ruter R2 (Hub). R2 je povezan sa R4 (ISP ruterom) koji vodi do Web Servera.

2. Konfiguracijski Zadaci

2.1 VLAN i STP

- Centrala: VLAN 10 (Prodaja: 192.168.10.0/24), VLAN 20 (Marketing: 192.168.20.0/24).
- Filijala: VLAN 30 (Gosti/Filijala: 192.168.30.0/24).
- Na S1 podesiti STP tako da on bude Root Bridge za VLAN 10 i 20. Svi međusobni linkovi između svičeva i prema ruterima su Trunk.

2.2 DHCP (R1 i R3)

- R1 dodjeljuje adrese za VLAN 10 i 20 (isključiti prvih 5 adresa). R3 dodjeljuje adrese za VLAN 30.

2.3 IP Adresiranje i RIPv2 Rutiranje

- WAN linkovi: R1-R2 (10.0.12.0/30), R3-R2 (10.0.23.0/30). Podinterfejsi na ruterima dobijaju prvu adresu (.1).
- Aktivirati RIP verziju 2 na ruterima R1, R2 i R3. Oglasiti sve privatne WAN i LAN mreže. Isključiti auto-summary.

2.4 Default Ruta i NAT/PAT (na R2 Hub-u)

- Veza prema ISP-u: R2-R4 (209.165.200.0/30). R2 dobija .1, R4 dobija .2.
- Na R2 kreirati default rutu prema R4 i proslijediti je kroz RIP ostalim ruterima.
- Na R2 podesiti PAT (Overload) tako da se kompletna 192.168.0.0/16 mreža prevodi u adresu Gig0/2 interfejsa prilikom izlaska na internet.

2.5 ACL - Sigurnost

- Kreirati Extended ACL na ruteru R3 i zabraniti Filijali (VLAN 30) pristup Prodaji (VLAN 10), ali im dozvoliti izlazak na internet (Server 8.8.8.8).

3. Rješenja (Konfiguracija)

HQ Sviči (S1, S2, S3) - Primjer za S1 i S2

```
! S1 (Core / Root Bridge)
enable
configure terminal
vlan 10
vlan 20
exit
interface range FastEthernet0/1 - 2, FastEthernet0/24
  switchport mode trunk
exit
spanning-tree vlan 10 root primary
spanning-tree vlan 20 root primary

! S2 (Access za VLAN 10)
enable
configure terminal
vlan 10
vlan 20
exit
interface range FastEthernet0/1, FastEthernet0/3
  switchport mode trunk
exit
interface FastEthernet0/10
  switchport mode access
  switchport access vlan 10
exit
```

Ruter R1 (HQ) - Interfejsi, DHCP, RIP

```
enable
configure terminal
interface GigabitEthernet0/0
  no shutdown
exit
interface GigabitEthernet0/0.10
  encapsulation dot1Q 10
  ip address 192.168.10.1 255.255.255.0
exit
interface GigabitEthernet0/0.20
  encapsulation dot1Q 20
  ip address 192.168.20.1 255.255.255.0
exit

ip dhcp excluded-address 192.168.10.1 192.168.10.5
ip dhcp pool PRODAJA
  network 192.168.10.0 255.255.255.0
  default-router 192.168.10.1
exit
ip dhcp excluded-address 192.168.20.1 192.168.20.5
ip dhcp pool MARKETING
  network 192.168.20.0 255.255.255.0
  default-router 192.168.20.1
exit

interface GigabitEthernet0/1
  ip address 10.0.12.1 255.255.255.252
  no shutdown
exit

router rip
  version 2
  no auto-summary
  network 192.168.10.0
  network 192.168.20.0
  network 10.0.0.0
exit
```

Ruter R2 (Hub) - RIP, Default Ruta i NAT

```
enable
configure terminal
interface GigabitEthernet0/0
 ip address 10.0.12.2 255.255.255.252
 ip nat inside
 no shutdown
exit
interface GigabitEthernet0/1
 ip address 10.0.23.1 255.255.255.252
 ip nat inside
 no shutdown
exit
interface GigabitEthernet0/2
 ip address 209.165.200.1 255.255.255.252
 ip nat outside
 no shutdown
exit

ip route 0.0.0.0 0.0.0.0 209.165.200.2

router rip
 version 2
 no auto-summary
 network 10.0.0.0
 default-information originate
exit

access-list 1 permit 192.168.0.0 0.0.255.255
ip nat inside source list 1 interface GigabitEthernet0/2 overload
```

Ruter R3 (Branch) - DHCP, RIP i ACL

```
enable
configure terminal
interface GigabitEthernet0/2
  no shutdown
exit
interface GigabitEthernet0/2.30
  encapsulation dot1Q 30
  ip address 192.168.30.1 255.255.255.0
exit

ip dhcp excluded-address 192.168.30.1 192.168.30.5
ip dhcp pool FILIJALA
  network 192.168.30.0 255.255.255.0
  default-router 192.168.30.1
exit

interface GigabitEthernet0/0
  ip address 10.0.23.2 255.255.255.252
  no shutdown
exit

router rip
  version 2
  no auto-summary
  network 192.168.30.0
  network 10.0.0.0
exit

ip access-list extended ZABRANI_HQ
  deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
  permit ip any any
exit
interface GigabitEthernet0/2.30
  ip access-group ZABRANI_HQ in
exit
```